# FINAL PROJECT REPORT

## Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

### 1. Abstract

This project focuses on automating the process of importing large datasets into the system and establishing relationships between tables using Import Sets and Dot Walking. Manual data entry is time-consuming and error-prone. By leveraging automated import mechanisms and relational mapping techniques, this project improves efficiency, accuracy, and scalability in data handling.

### 2. Introduction

Organizations deal with large volumes of data that need to be imported, transformed, and linked across multiple tables. Traditional methods require manual intervention, which increases the risk of inconsistencies.

This project provides a solution using:

- Import Sets for bulk data import
- Transform Maps for data conversion
- Dot Walking for relationship mapping between tables

### 3. Problem Statement

Manual data import and relationship creation:

- Takes too much time
- Leads to data inconsistencies
- Requires repetitive effort
- Lacks scalability

## 4. Objectives

- Automate bulk data import process
- Reduce manual errors
- Establish relationships between tables efficiently
- Improve data consistency
- Enhance system performance

## 5. System Overview

### 5.1 Modules

1. Data Import Module
2. Transformation Module
3. Relationship Mapping Module
4. User Interface Module (Login + Dashboard)

## 6. Technologies Used

| Category | Technology |
|---|---|
| Frontend | HTML, CSS, JavaScript |
| Backend | ServiceNow Platform |
| Database | ServiceNow Tables |
| Tools | Import Sets, Transform Maps |

## 7. Methodology

### Step 1: Data Upload

- Upload data (Excel/CSV) into Import Set table

### Step 2: Data Transformation

- Use Transform Map to map fields
- Apply business rules

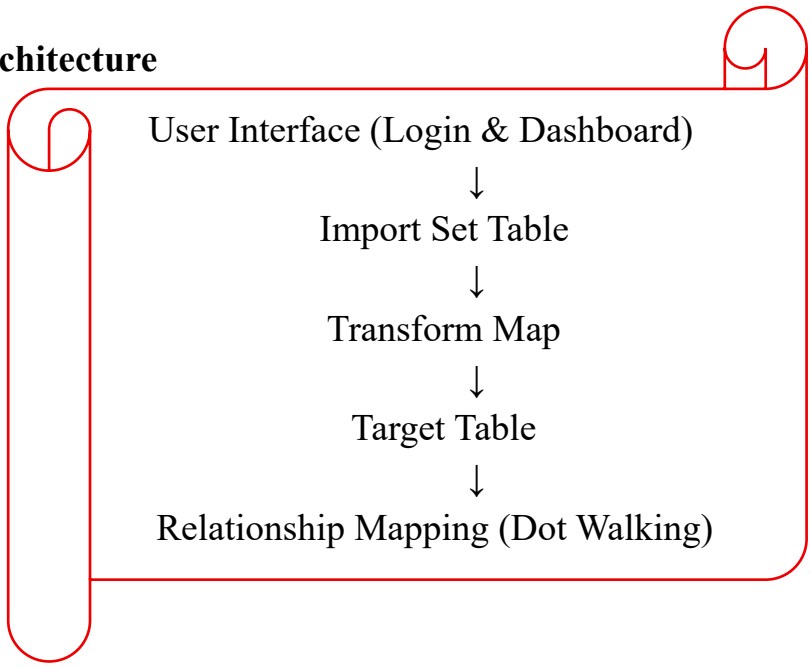
### Step 3: Data Insertion

- Insert transformed data into target tables

### Step 4: Relationship Mapping

- Use Dot Walking to link related records
- Example: Employee → Manager → Department

## 8. System Architecture




## 9. Features

- Secure Login System
- Dashboard with user details:

    - Name
    - Department
    - Phone Number
    - Manager
    - Email
- Automated data import
- Real-time data mapping
- Improved UI/UX

---

## 10. Implementation

### 10.1 Login Module

- Username & Password validation
- Redirect to dashboard

### 10.2 Dashboard

- Displays imported data
- Shows relationships using dot walking

### 10.3 Data Import

- Upload file into Import Set
- Run Transform Map

### 10.4 Relationship Mapping

- Use reference fields
- Access related data via dot walking

## 1. Ideation Phase

### 1.1 Problem Identification

Organizations face challenges in handling large volumes of data manually. Data import and relationship mapping are time-consuming and prone to errors.

### 1.2 Customer Problem Statement

| Customer | Need | Problem | Reason | Impact |
|---|---|---|---|---|
| **Admin/User** | Import bulk data | Manual entry is slow | No automation | Time-consuming & errors |

### 1.3 Empathy Understanding

- Users feel frustrated with repetitive tasks
- Fear of data inconsistency
- Need faster and reliable systems

## 2. Requirement Analysis Phase

### 2.1 Functional Requirements

- Upload data using Import Sets
- Transform data using Transform Maps
- Store data in target tables
- Display records in dashboard
- Map relationships using Dot Walking

### 2.2 Non-Functional Requirements

- Performance: Fast data processing
- Security: Login authentication

- Usability: Simple UI
- Scalability: Handle large datasets

## 3. Project Design Phase

### 3.1 System Architecture

User → Login Page → Dashboard → Import Set → Transform Map → Target Table → Dot Walking

### 3.2 Database Design

Tables Used:

- Import Set Table
- Target Table (Employee Data)

Fields:

- Name
- Department
- Phone Number
- Manager (Reference Field)
- Email

### 3.3 UI Design

- Login Page
- Dashboard Page
- Data Display Table

## 4. Project Planning Phase

### 4.1 Timeline

| Phase | Duration |
|---|---|
| **Ideation** | 1 Day |
| **Requirement Analysis** | 1 Day |
| **Design** | 2 Days |
| **Development** | 3 Days |
| **Testing** | 1 Day |

### 4.2 Resource Planning

- System with browser
- ServiceNow platform
- Dataset (CSV/Excel)

## 5. Project Development Phase

### 5.1 Implementation Steps

**Step 1: Create Import Set**

- Load Excel/CSV data

**Step 2: Create Transform Map**

- Map source fields to target fields

**Step 3: Run Transformation**

- Insert data into target table

**Step 4: Apply Dot Walking**

- Link Manager → Employee

- Access related data

**Step 5: UI Development**

- Login page with authentication
- Dashboard displaying data

## 6. Project Testing Phase

### 6.1 Test Cases

| Test Case | Expected Result |
|---|---|
| Login | Successful access |
| Data Upload | Data imported correctly |
| Mapping | Correct field mapping |
| Dot Walking | Relationships displayed |

### 6.2 Results

- All modules working correctly
- No major errors found

## 7. Project Documentation Phase

### 7.1 User Guide

1. Login using credentials
2. Upload dataset
3. Run transform map
4. View data in dashboard

### 7.2 Technical Documentation

- Import Set configuration

- Transform Map setup
- Reference field mapping

### 8. Project Demonstration Phase

#### 8.1 Demo Flow

1. Login into system
2. Upload sample dataset
3. Execute transformation
4. Display imported data
5. Show relationship mapping

#### 8.2 Output

- Data successfully imported
- Relationships correctly mapped
- Dashboard displays complete information

## 11. Advantages

- Saves time
- Reduces manual errors
- Ensures data consistency
- Scalable for large datasets
- Improves productivity

## 12. Limitations

- Requires initial setup
- Depends on correct mapping
- Needs knowledge of ServiceNow

## 13. Results

- Successfully automated data import
- Achieved accurate relationship mapping
- Improved system efficiency
- Reduced manual workload

## 14. Future Enhancements

- Real-time data integration
- API-based import automation
- Advanced analytics dashboard
- AI-based error detection

## 16. References


- ServiceNow Documentation
- Online Tutorials on Import Sets
- Database Management Concepts


## 17. Appendix

**Sample Fields Used:**

- Name
- Department
- Phone Number
- Manager
- Email

## 15. Conclusion

The project successfully demonstrates how automation using Import Sets and Dot Walking can simplify complex data operations. It significantly improves efficiency, reduces errors, and ensures accurate relationship mapping. This approach is highly beneficial for organizations managing large-scale data systems.